

Most projects were created in JupyterHub, with other tools incorporated. Additional information is available within each of the assignment folders.

Project A:

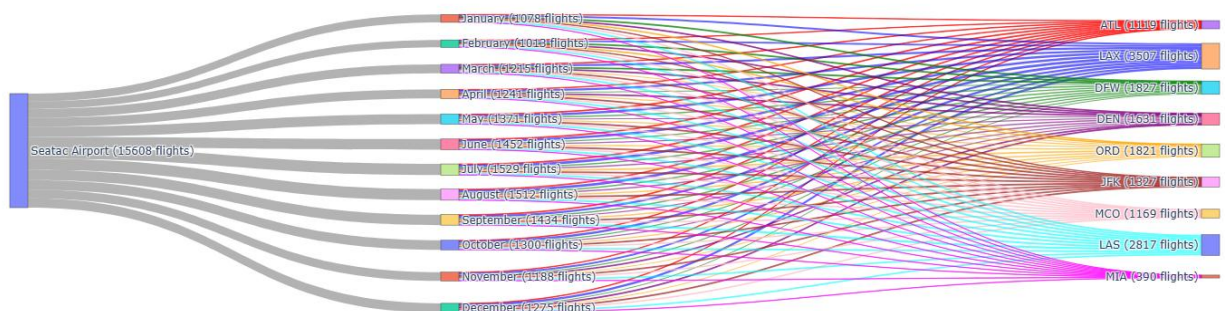
For this first project, we were asked to create a diagram representing consistent movement across Seattle. The diagram in question was a Sankey Diagram, a helpful visual that displays the direction and quantity of connections between multiple modes. Since Sea-Tac International Airport shares close proximity to Seattle and is a prominent location in the area, my team elected to display flights moving through Sea-Tac.

This project had a two-week turnaround with four group workdays, so we had to manage the scope of our diagram. We chose to plot outbound flights from Sea-Tac International Airport to the 9 most-common destination airports in the US, taking data from January through December of 2024. We further refined our parameters by looking solely at flights run by Alaska Airlines, one of the most prominent airlines operating from Sea-Tac. We were interested in seeing the number of flights-per-month taking off from Sea-Tac, and where those flights were headed. We chose the following destination airports: We chose the following: Atlanta ATL; Denver DEN; Dallas DFW; New York City JFK; Las Vegas LAS; Los Angeles LAX; Orlando MCO; Miami MIA; Chicago ORD. Our data was pulled from the Bureau of Transportation Statistics website, with the portal available here: <https://www.transtats.bts.gov/ontime/departures.aspx>.

A normal Sankey diagram can have multiple starting nodes and multiple layers of nodes later on. Our diagram is more standard, with a single starting node at Sea-Tac and two additional layers of nodes, representing the month of the year and the destination airport.

Below is our Sankey Diagram.

Flow of Alaska Airlines flights each month from Seatac airport to destination airport in 2024



At a broad glance, the diagram shows that Alaska Airlines operated out of Seattle for all 12 months of 2024, and flew to all nine of our destination airports throughout those 12 months. This isn't surprising, but shows that these flight routes are very consistent and well traveled.

Predictably, the summer months show the highest amount of traffic, with over 1400 flights a month consequently from June through September. In Washington state this aligns with summer holidays for public school and college students. January and February had the least number of flights with under 1100 flights, with higher volume in the Spring and Autumn. I found it interesting that October had more flights recorded than November and December, since the latter two months both have major holidays that normally offer time off from work and school (Thanksgiving in November, the last two weeks of December with Christmas in particular).

Looking at the destinations from Sea-Tac, Los Angeles LAX and Las Vegas LAS were the clear favorites. Out of the total nine destinations, six of them lie in the south of the country: LAX, LAS, DFW, ATL, MIA, and MCO. The remaining three (DEN, ORD, and JFK) sit in the central and northeast of the nation.

Engineer: to create like He did

Future changes or expansions?

Project B: